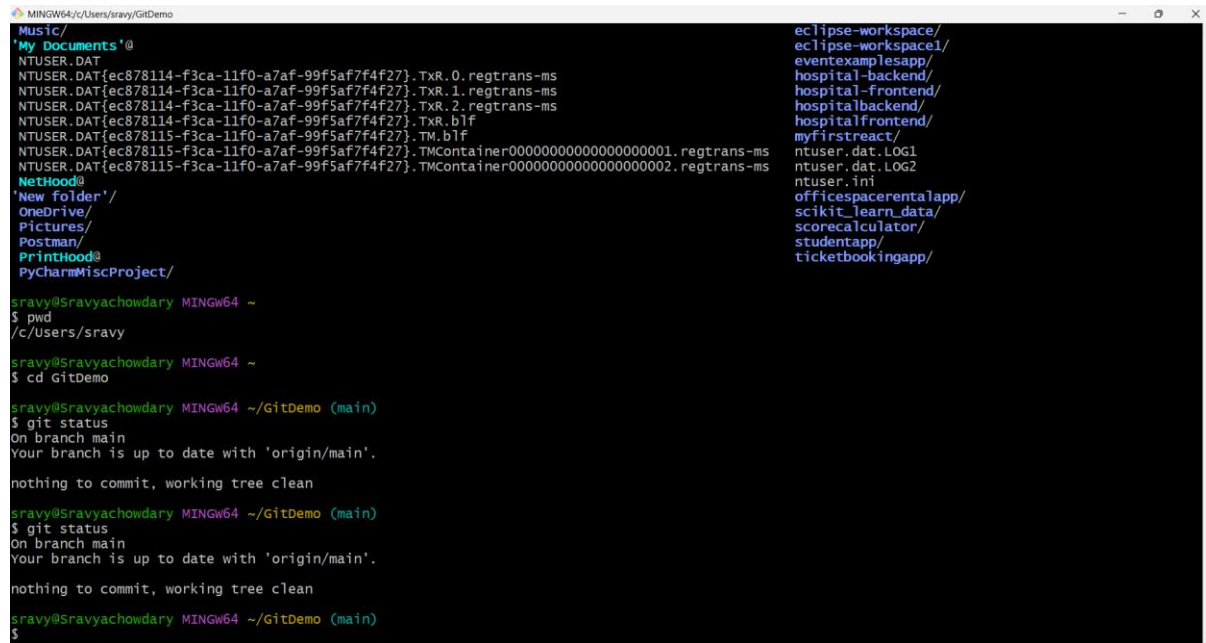


GIT

4. Git-HOL

Step 1: Verify master is clean



```

MINGW64~/c/Users/sravy/GitDemo
Music/
'My Documents'@
NTUSER.DAT
NTUSER.DAT{ec878114-f3ca-11f0-a7af-99f5af7f4f27}.TxR.0.regtrans-ms
NTUSER.DAT{ec878114-f3ca-11f0-a7af-99f5af7f4f27}.TxR.1.regtrans-ms
NTUSER.DAT{ec878114-f3ca-11f0-a7af-99f5af7f4f27}.TxR.2.regtrans-ms
NTUSER.DAT{ec878114-f3ca-11f0-a7af-99f5af7f4f27}.TxR.b1f
NTUSER.DAT{ec878115-f3ca-11f0-a7af-99f5af7f4f27}.TM.b1f
NTUSER.DAT{ec878115-f3ca-11f0-a7af-99f5af7f4f27}.TMcontainer00000000000000000001.regtrans-ms
NTUSER.DAT{ec878115-f3ca-11f0-a7af-99f5af7f4f27}.TMcontainer00000000000000000002.regtrans-ms
NetHood@
'New Folder'/'
OneDrive/
Pictures/
Postman/
PrintHood@
PyCharmMiscProject/

savy@Savyachowdary MINGW64 ~
$ pwd
/c/Users/sravy

savy@Savyachowdary MINGW64 ~
$ cd GitDemo

savy@Savyachowdary MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

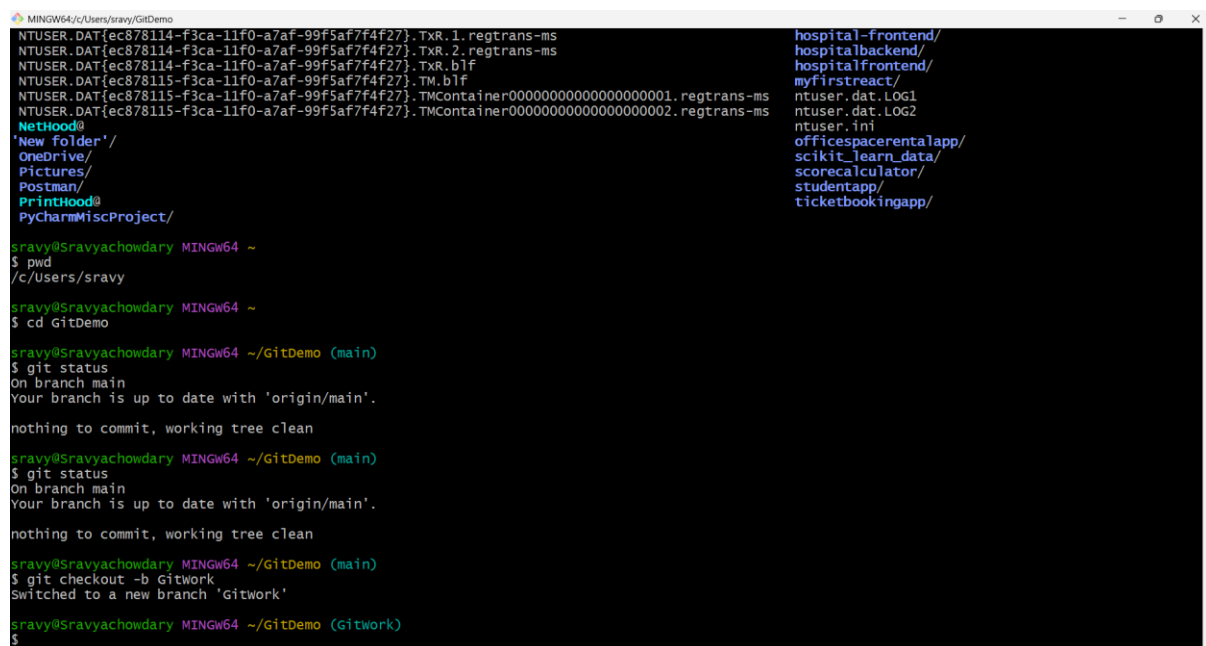
nothing to commit, working tree clean

savy@Savyachowdary MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

savy@Savyachowdary MINGW64 ~/GitDemo (main)
$
  
```

Step 2: Create branch



```

MINGW64~/c/Users/sravy/GitDemo
NTUSER.DAT{ec878114-f3ca-11f0-a7af-99f5af7f4f27}.TxR.1.regtrans-ms
NTUSER.DAT{ec878114-f3ca-11f0-a7af-99f5af7f4f27}.TxR.2.regtrans-ms
NTUSER.DAT{ec878114-f3ca-11f0-a7af-99f5af7f4f27}.TxR.b1f
NTUSER.DAT{ec878115-f3ca-11f0-a7af-99f5af7f4f27}.TM.b1f
NTUSER.DAT{ec878115-f3ca-11f0-a7af-99f5af7f4f27}.TMcontainer00000000000000000001.regtrans-ms
NTUSER.DAT{ec878115-f3ca-11f0-a7af-99f5af7f4f27}.TMcontainer00000000000000000002.regtrans-ms
NetHood@
'New Folder'/'
OneDrive/
Pictures/
Postman/
PrintHood@
PyCharmMiscProject/

savy@Savyachowdary MINGW64 ~
$ pwd
/c/Users/sravy

savy@Savyachowdary MINGW64 ~
$ cd GitDemo

savy@Savyachowdary MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

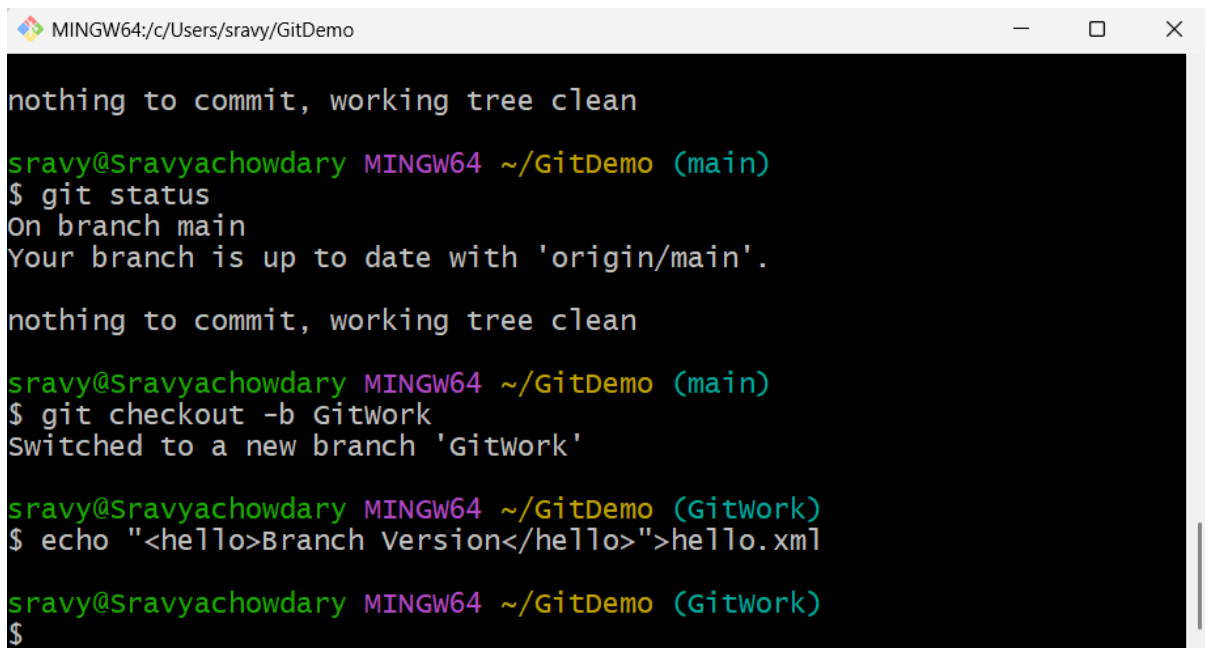
savy@Savyachowdary MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

savy@Savyachowdary MINGW64 ~/GitDemo (main)
$ git checkout -b Gitwork
Switched to a new branch 'Gitwork'

savy@Savyachowdary MINGW64 ~/GitDemo (Gitwork)
$
  
```

Step 3: Create hello.xml

A terminal window titled 'MINGW64:/c/Users/sravy/GitDemo' with standard window controls. The terminal shows the following commands and output:

```
nothing to commit, working tree clean

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

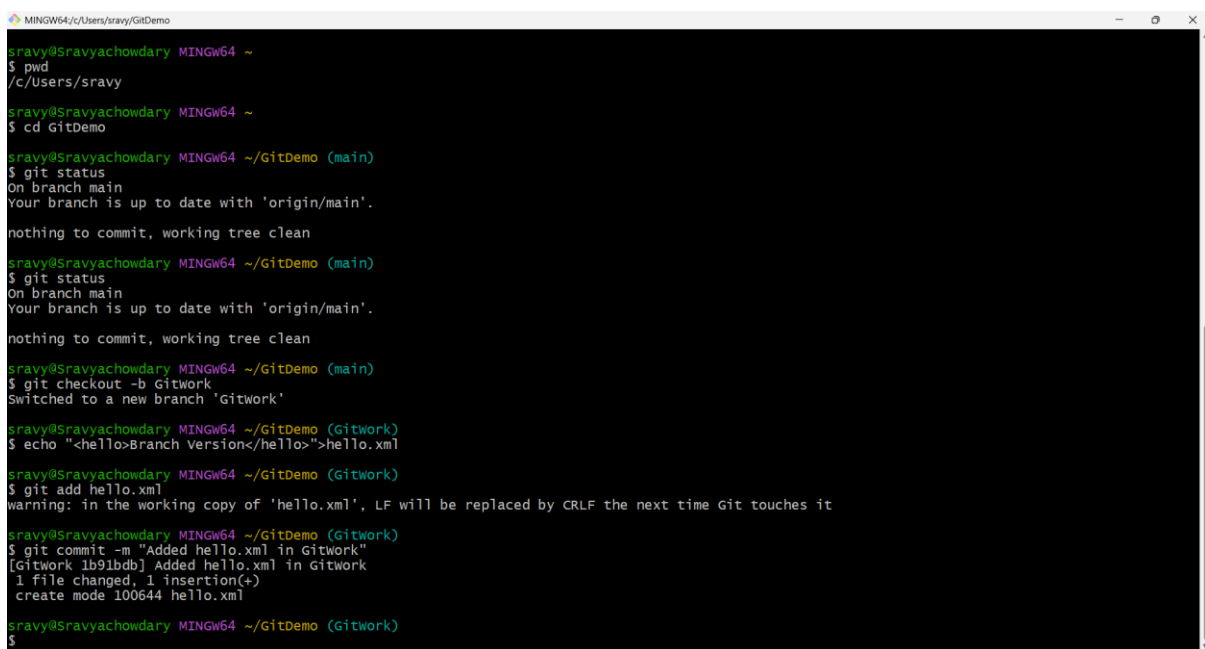
nothing to commit, working tree clean

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git checkout -b Gitwork
Switched to a new branch 'Gitwork'

sravy@Sravyachowdary MINGW64 ~/GitDemo (Gitwork)
$ echo "<hello>Branch Version</hello>">hello.xml

sravy@Sravyachowdary MINGW64 ~/GitDemo (Gitwork)
$
```

Step 4: Commit

A terminal window titled 'MINGW64:/c/Users/sravy/GitDemo' with standard window controls. The terminal shows the following commands and output:

```
sravy@Sravyachowdary MINGW64 ~
$ pwd
/c/Users/sravy

sravy@Sravyachowdary MINGW64 ~
$ cd GitDemo

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git checkout -b Gitwork
Switched to a new branch 'Gitwork'

sravy@Sravyachowdary MINGW64 ~/GitDemo (Gitwork)
$ echo "<hello>Branch Version</hello>">hello.xml

sravy@Sravyachowdary MINGW64 ~/GitDemo (Gitwork)
$ git add hello.xml
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it

sravy@Sravyachowdary MINGW64 ~/GitDemo (Gitwork)
$ git commit -m "Added hello.xml in Gitwork"
[Gitwork 1b91bdb] Added hello.xml in Gitwork
1 file changed, 1 insertion(+)
create mode 100644 hello.xml

sravy@Sravyachowdary MINGW64 ~/GitDemo (Gitwork)
$
```

Step 5: Switched from GitWork branch to Main branch

```
sravy@Sravyachowdary MINGW64 ~/GitDemo (Gitwork)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$
```

Step 6: Create hello.xml in the main branch with different content

```
MINGW64:/c/Users/sravy/GitDemo
$ git commit -m "Added hello.xml in Gitwork"
[Gitwork 1b91bdb] Added hello.xml in Gitwork
1 file changed, 1 insertion(+)
create mode 100644 hello.xml

sravy@Sravyachowdary MINGW64 ~/GitDemo (GitWork)
$ git checkout master
error: pathspec 'master' did not match any file(s) known to git

sravy@Sravyachowdary MINGW64 ~/GitDemo (GitWork)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ echo "<hello>Main Version</hello>" > hello.xml

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$
```

Step 7: Committed hello.xml changes in Main branch

```
MINGW64:/c/Users/sravy/GitDemo
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ echo "<hello>Main Version</hello>" > hello.xml

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git add hello.xml
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next
time Git touches it

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git commit -m "Added hello.xml in main"
[main 887881d] Added hello.xml in main
1 file changed, 1 insertion(+)
create mode 100644 hello.xml

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$
```

Step 8: Git Commit History

```

MINGW64/c/Users/sravy/GitDemo
$ git commit -m "Added hello.xml in Gitwork"
[Gitwork 1b91bdb] Added hello.xml in Gitwork
1 file changed, 1 insertion(+)
create mode 100644 hello.xml

sravy@Sravyachowdary MINGW64 ~/GitDemo (Gitwork)
$ git checkout master
error: pathspec 'master' did not match any file(s) known to git

sravy@Sravyachowdary MINGW64 ~/GitDemo (Gitwork)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ echo "<hello>Main Version</hello>" > hello.xml

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git add hello.xml
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git commit -m "Added hello.xml in main"
[main 887881d] Added hello.xml in main
1 file changed, 1 insertion(+)
create mode 100644 hello.xml

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git log --oneline --graph --decorate --all
* 887881d (HEAD -> main) Added hello.xml in main
| * 1b91bdb (Gitwork) Added hello.xml in Gitwork
|/
* d7ca269 (origin/main) Merge branch 'main' of https://gitlab.com/sravyakilaru68/gitdemo
| \
| * 3e344be Initial commit
* 0a6d3e3 Added gitignore for log files
* 78f6a74 Added branch file
* 4be129d First

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$

```

Step 9: Difference between Main and GitWork Branches

```

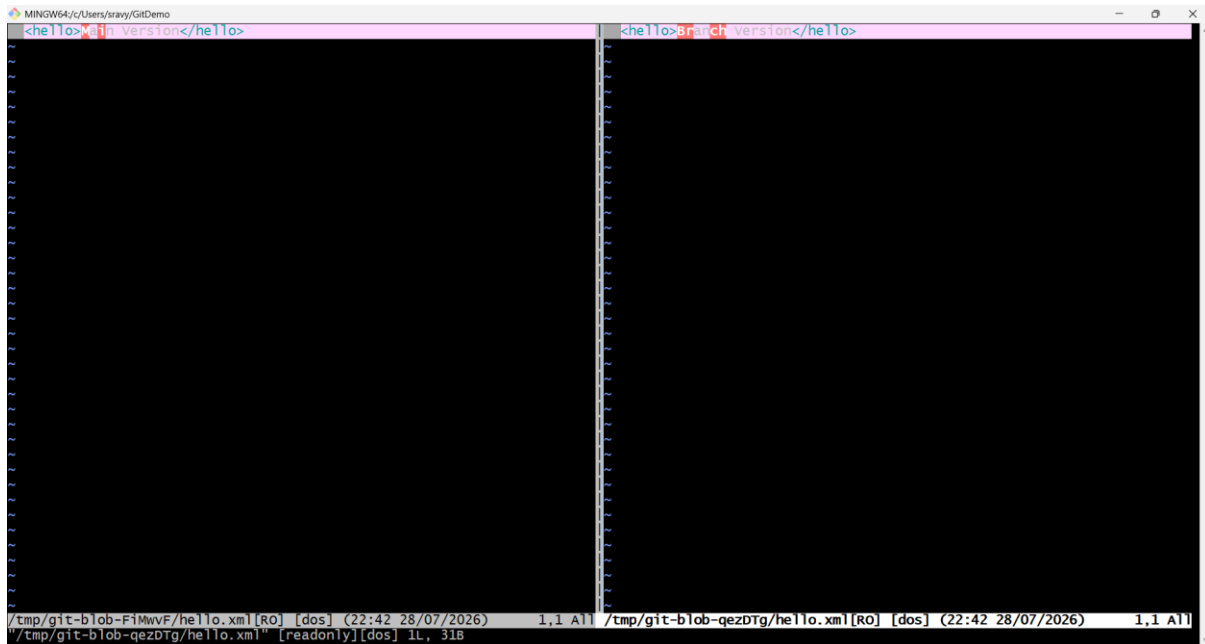
sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git diff main Gitwork
diff --git a/hello.xml b/hello.xml
index 2ee562d..f12c303 100644
--- a/hello.xml
+++ b/hello.xml
@@ -1,1 @@
-<hello>Main Version</hello>
+<hello>Branch Version</hello>

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$

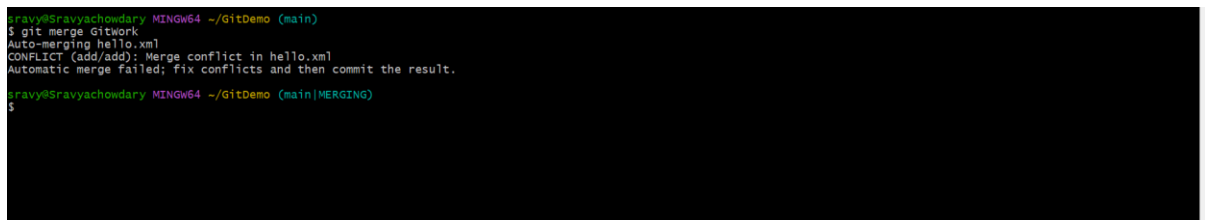
```

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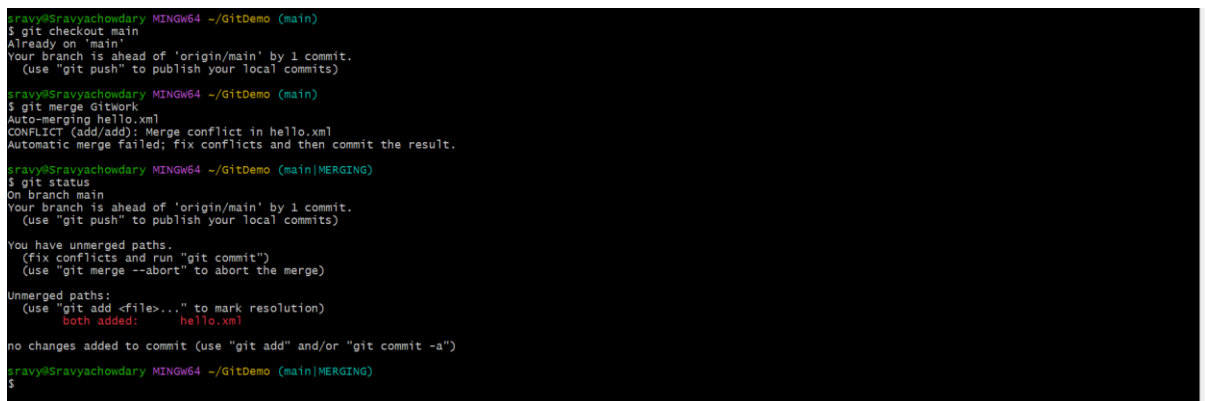
Step 10: Visual Comparison of Main and GitWork Branches Using Diff Tool (P4Merge/Git Difftool)



Step 11: Merge the branch



Step 12: Git Status After Merge Conflict



Step 13: Resolved Merge Conflict

```
pravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git checkout main
Already on 'main'
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

pravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git merge Gitwork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

pravy@Sravyachowdary MINGW64 ~/GitDemo (main|MERGING)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

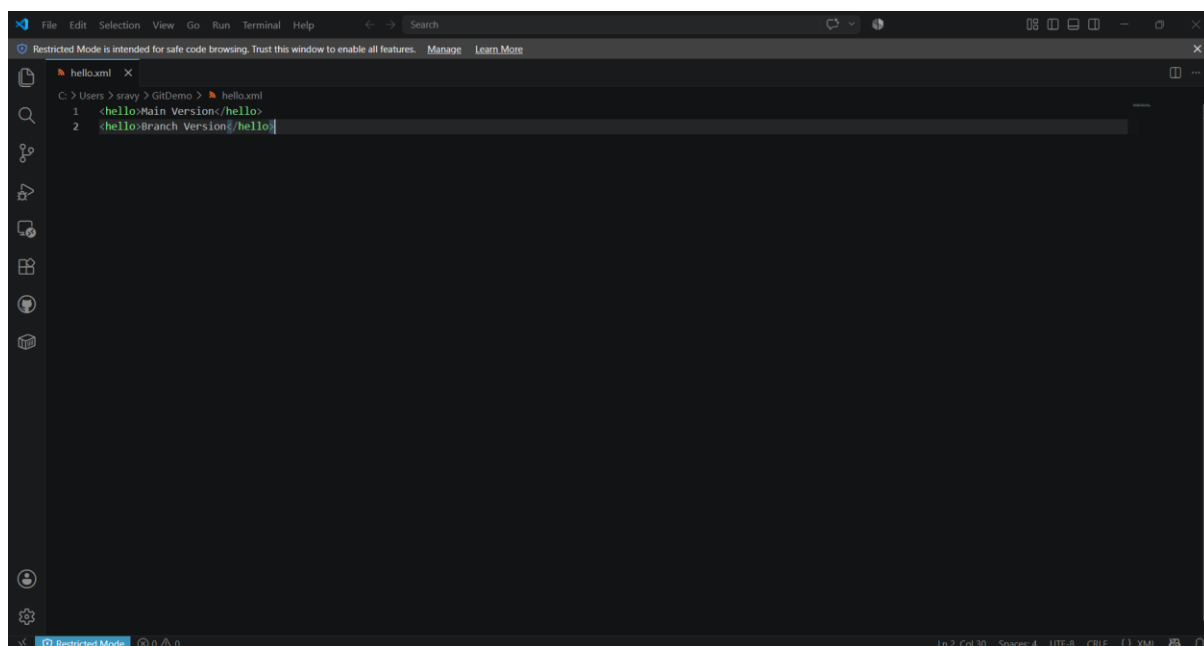
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both added:   hello.xml

no changes added to commit (use "git add" and/or "git commit -a")

pravy@Sravyachowdary MINGW64 ~/GitDemo (main|MERGING)
$ code hello.xml

pravy@Sravyachowdary MINGW64 ~/GitDemo (main|MERGING)
$
```



Step 14: Commit the resolved file

```
pravy@Sravyachowdary MINGW64 ~/GitDemo (main|MERGING)
$ code hello.xml

pravy@Sravyachowdary MINGW64 ~/GitDemo (main|MERGING)
$ git add hello.xml

pravy@Sravyachowdary MINGW64 ~/GitDemo (main|MERGING)
$ git commit -m "Resolved merge conflict"
[main ba9bb05] Resolved merge conflict

pravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$
```

Step 15: Check status

```
pravy@Sravyachowdary MINGW64 ~/GitDemo (main|MERGING)
$ git commit -m "Resolved merge conflict"
[main ba9bb05] Resolved merge conflict

pravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 3 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

pravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$
```

Step 16: Update .gitignore

```
MINGW64:/c/Users/sravy/GitDemo
sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 3 commits.
(use "git push" to publish your local commits)

nothing to commit, working tree clean

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ echo "*.orig" >> .gitignore

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git add .gitignore
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git commit -m "Updated .gitignore"
[main 29c04d6] Updated .gitignore
1 file changed, 1 insertion(+), 1 deletion(-)

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ |
```

Step 17: List branches

```
MINGW64:/c/Users/sravy/GitDemo
nothing to commit, working tree clean

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ echo "*.orig" >> .gitignore

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git add .gitignore
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git commit -m "Updated .gitignore"
[main 29c04d6] Updated .gitignore
1 file changed, 1 insertion(+), 1 deletion(-)

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git branch
  GitWork
* main

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ |
```

Step 18: Delete the branch

```
MINGW64:/c/Users/sravy/GitDemo
$ echo "*.orig" >> .gitignore

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git add .gitignore
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git commit -m "Updated .gitignore"
[main 29c04d6] Updated .gitignore
1 file changed, 1 insertion(+), 1 deletion(-)

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git branch
  GitWork
* main

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git branch -d GitWork
Deleted branch GitWork (was 1b91bdb).

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ |
```

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Step 19: View the final log

```
MINGW64:/c/Users/sravy/GitDemo
[main 29c04d6] Updated .gitignore
1 file changed, 1 insertion(+), 1 deletion(-)

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git branch
* GitWork
* main

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git branch -d GitWork
Deleted branch GitWork (was 1b91bdb).

sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$ git log --oneline --graph --decorate
* 29c04d6 (HEAD -> main) Updated .gitignore
*   ba9bb05 Resolved merge conflict
|
| * 1b91bdb Added hello.xml in GitWork
| * | 887881d Added hello.xml in main
|/
* d7ca269 (origin/main) Merge branch 'main' of https://gitlab.com/sravyakilaru68/gitdemo
|
| * 3e344be Initial commit
| * 0a6d3e3 Added gitignore for log files
| * 78f6a74 Added branch file
| * 4be129d First
|
sravy@Sravyachowdary MINGW64 ~/GitDemo (main)
$
```